

Qatar-1b

R-2818

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_130723 · created 2026-09-05T02:13:26+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456496.807503 +/- 8.5e-05 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.189 +/- 0.056
Transit depth (Rp/Rs)^2	3.57 +/- 2.13 [%]
Orbital Inclination (inc)	82.35 +/- 2.88
Airmass coefficient 1 (a1)	0.18 +/- 0.11
Airmass coefficient 2 (a2)	1.39 +/- 1.74
Scatter in the residuals of the lightcurve fit is	59.35 %
Best Comparison Star	None
Optimal Aperture	2.17
Optimal Annulus	5.5
Transit Duration (day)	0.053 +/- 0.027

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.189 $\pm$ 0.056 vs 0.1463 (deviation 0.67 σ)
Duration fit vs published	0.053 d vs 0.0692 d (deviation 0.52 σ)
Mid-time O-C	-0.2 min
Depth SNR	3.0
Residual scatter	59.35 %

Light curve

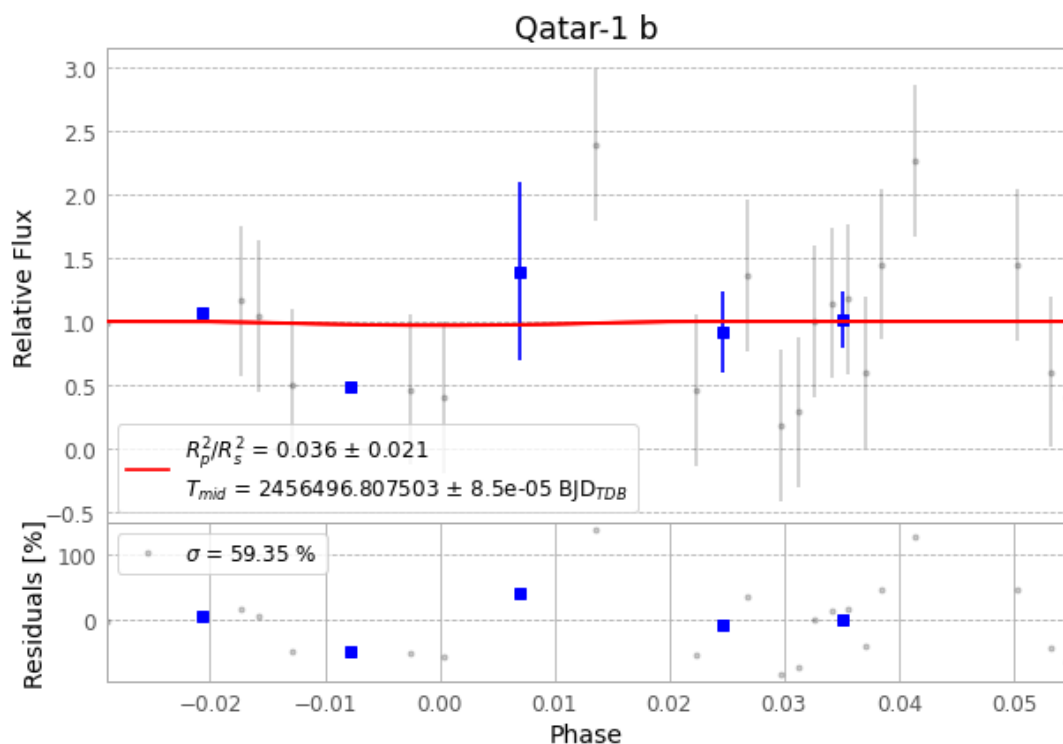


Figure 1 — FinalLightCurve_Qatar-1 b_2013-07-22.png (SHA-256 8ca8d5bb16e0e675...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [201, 329], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 bb85d133b6b4bb6c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

60 FITS frames (39 MB), Qatar-1130723062112.FITS → Qatar-1130723091812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-130723.log (a688619ff7ed...), FinalLightCurve_Qatar-1 b_2013-07-22.png (8ca8d5bb16e0...), FinalLightCurve_Qatar-1 b_2013-07-22.csv (01f87eac2f01...)

Compute resources

Wall time 466.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-05 02:13Z.